

SAIGO, Japan

WMO: 477400

Lat: 36.2044N

Lon: 133.3339E

Elev: 92

StdP: 14.65

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	28.9	30.3	19.0	14.3	35.6	21.2	15.9	37.0	24.5	40.4	22.0	41.7	6.8	310	0.438

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.1	88.8	78.1	86.9	77.5	85.1	76.9	79.6	85.8	78.7	84.8	77.7	83.6	9.0	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.9	145.2	83.7	76.9	140.3	82.9	75.8	135.5	82.1	43.3	85.8	42.3	84.9	41.3	83.9	84.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
21.3	18.8	16.5	DB	25.4	91.7	2.5	1.9	23.6	93.1	22.2	94.2	20.8	95.3	19.0	96.7
			WB	23.6	81.1	2.4	1.5	21.9	82.2	20.5	83.1	19.1	83.9	17.4	85.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	58.2	39.8	40.1	44.6	53.2	61.8	68.8	76.8	79.6	72.2	62.6	53.3	44.6
	DBStd	14.46	4.49	4.88	5.63	5.81	4.52	4.03	4.43	3.74	4.59	4.87	5.38	5.38
	HDD50	1022	318	281	185	31	0	0	0	0	0	0	27	180
	HDD65	3683	782	698	632	354	119	10	0	0	2	103	351	632
	CDD50	4023	1	3	19	128	366	563	830	917	666	390	127	13
	CDD65	1206	0	0	0	1	19	123	365	452	218	28	0	0
	CDH74	8769	0	0	0	3	66	382	2769	4356	1141	52	0	0
	CDH80	2306	0	0	0	0	3	33	681	1397	189	3	0	0
Wind	WSAvg	7.4	8.0	7.6	7.9	7.9	7.1	6.5	7.3	6.9	7.0	7.0	7.7	8.2
Precipitation	PrecAvg	71.9	6.7	4.4	4.9	4.6	5.6	6.5	8.3	5.6	8.8	4.7	5.4	6.4
	PrecMax	94.6	12.2	7.7	7.8	9.7	11.1	17.8	21.9	15.2	15.4	10.0	8.6	12.0
	PrecMin	53.3	3.8	2.4	1.8	1.0	2.2	1.3	0.1	0.8	1.6	2.0	2.7	3.5
	PrecStd	8.5	1.9	1.3	1.5	2.1	2.7	3.7	5.1	4.1	3.9	2.2	1.7	1.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.8	60.2	65.5	73.4	79.0	83.6	90.2	91.6	87.3	78.2	70.4	62.0
		MCWB	51.6	54.0	54.9	59.3	63.9	71.3	77.8	78.7	77.1	70.7	62.7	56.0
	2%	DB	53.1	55.6	61.0	69.2	75.7	80.3	87.9	89.6	84.5	75.3	67.5	58.8
		MCWB	46.8	50.3	52.5	57.8	63.3	70.3	77.6	78.4	75.9	68.2	60.6	52.6
	5%	DB	50.6	52.2	58.2	66.5	73.3	78.1	85.9	88.0	82.0	73.2	65.6	56.4
		MCWB	44.4	46.6	51.4	57.1	61.9	70.0	76.8	78.2	74.5	65.7	59.0	49.9
	10%	DB	48.2	49.5	55.7	63.8	71.1	76.0	83.8	86.3	79.8	71.3	63.5	54.2
		MCWB	42.7	44.1	49.5	55.8	61.2	69.1	76.1	77.7	73.0	64.2	57.2	48.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.2	55.3	57.2	64.1	69.0	74.8	79.8	81.4	78.8	72.9	64.8	58.1
		MCDB	55.6	59.5	61.4	68.3	72.9	79.9	86.3	87.8	85.0	77.0	67.8	60.3
	2%	WB	48.3	50.9	54.5	60.6	66.4	73.4	78.6	80.1	77.1	70.2	62.3	53.7
		MCDB	52.4	54.3	59.0	66.0	71.0	77.5	85.1	86.4	82.6	73.9	66.4	57.6
	5%	WB	45.4	47.4	52.4	58.7	64.7	71.8	77.7	79.3	75.7	67.6	60.2	51.0
		MCDB	49.3	51.3	57.2	64.0	70.3	76.0	83.8	85.4	80.7	71.3	64.5	55.3
	10%	WB	43.2	44.7	50.1	56.9	63.1	70.3	76.8	78.6	74.2	65.5	57.9	48.5
		MCDB	47.6	49.0	55.0	62.2	68.9	74.5	82.6	84.5	78.8	69.9	62.6	53.2
Mean Daily Temperature Range	5% DB	MDBR	11.0	12.6	14.9	15.7	14.6	11.8	9.8	11.1	11.7	13.8	13.4	11.8
		MCDBR	15.6	17.5	19.0	19.4	19.0	14.8	13.1	13.3	13.4	15.0	15.4	15.2
		MCWBR	11.5	13.0	13.3	11.3	10.3	7.6	5.8	5.7	6.9	9.0	10.5	11.0
	5% WB	MCDBR	14.3	16.1	16.8	15.4	14.6	10.5	11.9	12.0	11.4	12.1	13.7	14.1
		MCWBR	11.7	13.3	13.2	10.4	9.4	6.0	5.8	5.8	6.4	8.6	10.6	11.7
Clear-Sky Solar Irradiance	taub	0.448	0.480	0.548	0.562	0.561	0.576	0.575	0.527	0.484	0.459	0.441	0.442	
	taud	1.982	1.920	1.789	1.810	1.871	1.917	1.983	2.130	2.188	2.126	2.087	2.004	
	Ebn at Noon	222	232	228	235	238	233	233	242	245	238	224	214	
	Edh at Noon	44	52	65	67	64	61	57	48	43	42	39	40	
All-Sky Solar Radiation	RadAvg	532	788	1183	1548	1685	1623	1585	1652	1278	1031	678	475	
	RadStd	44	70	97	132	164	122	212	194	100	89	58	37	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.56	N/A	N/A	N/A	N/A

Nomenclature: See separate page